

Project Design Phase-II
Data Flow Diagram & User Stories

Date	25 June 2026
Team ID	Not Given
Project Name	Rising Waters
Maximum Marks	4 Marks

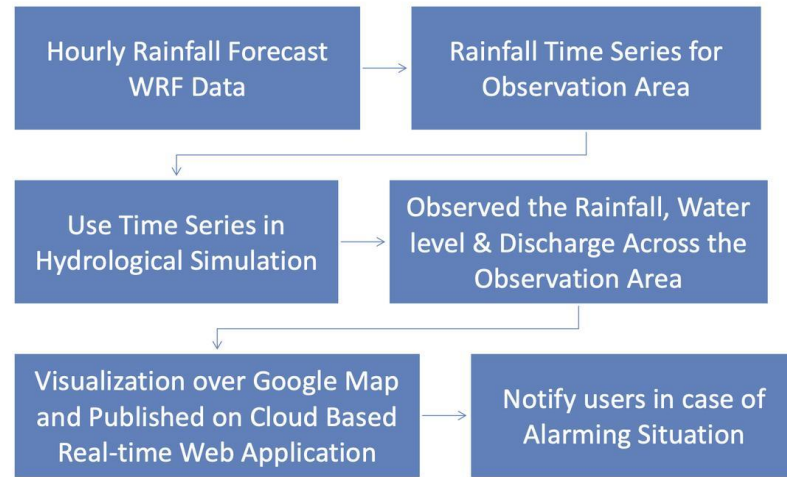
Team Members	Skill Wallet Team ID's	Roll Numbers (College: ACET)
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Data Flow Diagrams:

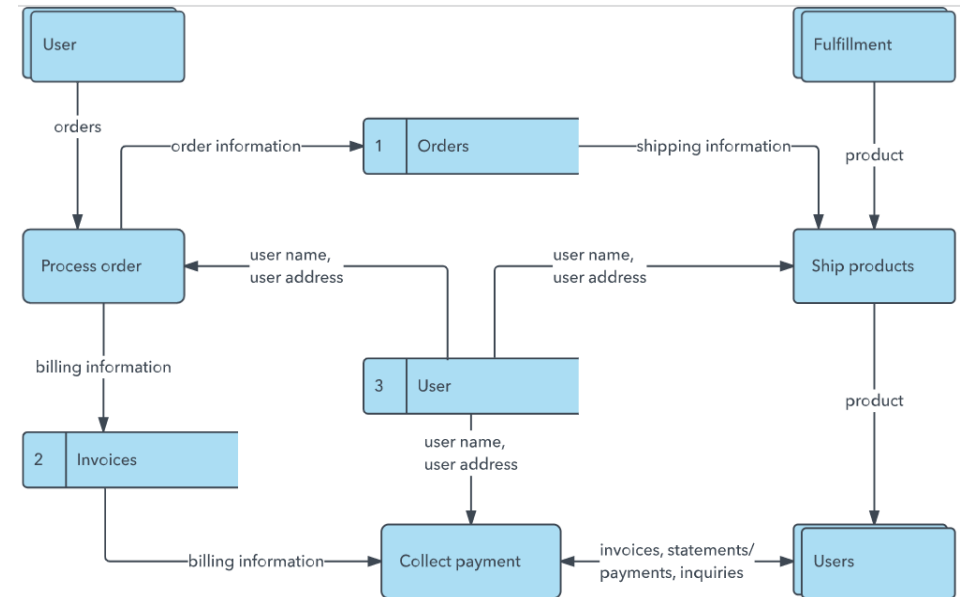
The Data Flow Diagram (DFD) illustrates how data moves through the **Rising Waters** flood prediction system. Weather parameters entered by users are validated and processed by the Flask application, passed to the trained machine learning model for prediction, and the resulting flood risk is displayed to users while prediction records are stored for future analysis and monitoring.

Example: [\(Simplified\)](#)

System Function



Example: DFD Level 0 (Industry Standard)



User enters weather parameters such as annual rainfall, cloud visibility, temperature, humidity, and seasonal rainfall through the web application. The Flask application validates the input data and prepares it for processing and prediction. Data preprocessing and feature scaling are performed using the saved scaler object to match the model training format. The processed data is sent to the trained XGBoost model to analyze flood risk and generate a prediction. The prediction result (Flood / No Flood) and probability score are displayed in the user interface and stored for future analysis and monitoring.

User Stories : Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Disaster Management Officer	Registration	USN-1	As a user, I can register by entering my name, email, password, and role.	User account is created successfully.	High	Sprint-1
Disaster Management Officer	Login	USN-2	As a user, I can log in securely using my email and password.	User can access the dashboard after successful login.	High	Sprint-1
Disaster Management Officer	Dashboard	USN-3	As a user, I can view the flood prediction dashboard.	Dashboard displays prediction options and history.	High	Sprint-1
Disaster Management Officer	Flood Prediction	USN-4	As a user, I can enter weather parameters to predict flood risk.	System displays Flood/No Flood result with probability.	High	Sprint-1
Meteorologist	Weather Data Entry	USN-5	As a meteorologist, I can input rainfall, temperature, humidity, cloud visibility, and seasonal rainfall data.	Input data is validated before prediction.	High	Sprint-1
Meteorologist	Prediction History	USN-6	As a meteorologist, I can view previous flood predictions.	Previous prediction records are displayed correctly.	Medium	Sprint-2
Government Analyst	Model Performance	USN-7	As an analyst, I can compare the accuracy of different ML models.	Accuracy and evaluation metrics are displayed.	Medium	Sprint-2
Government Analyst	Report Generation	USN-8	As an analyst, I can generate flood prediction reports.	Reports can be viewed or downloaded successfully.	Medium	Sprint-2
Administrator	User Management	USN-9	As an administrator, I can manage user accounts and roles.	Users can be added, updated, or removed.	High	Sprint-2
Administrator	Model Management	USN-10	As an administrator, I can upload or update the trained ML model.	The new model is successfully loaded into the system.	High	Sprint-2
Administrator	Prediction Monitoring	USN-11	As an administrator, I can monitor all prediction activities.	System logs and prediction records are available.	Medium	Sprint-2
Disaster Management Officer	Cloud Deployment	USN-12	As a user, I can access the application from any location through IBM Cloud.	Application is available online with secure access.	Low	Sprint-3